

Olasılıksal Robotik

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Haritalama

- Occupancy Grid Mapping (Bilinen Konum ile Haritalama)

$$p \left(m \middle| z_{1:t}, x_{1:t} \right)$$

$$m = \{\mathbf{m}_i\}$$

- $\mathbf{m}_i$ uzayı temsil eden hücreler

$$p(\mathbf{m}_i = 1) \implies \text{hücre dolu}$$

Haritalama

- Binary Bayes Filtresi gibi ele alınabilir

$$p\left(m \middle| z_{1:t}, x_{1:t}\right) = \prod_i p\left(\mathbf{m}_i \middle| z_{1:t}, x_{1:t}\right)$$

Occupancy Grid Mapping

```
1:  Algorithm occupancy_grid_mapping( $\{l_{t-1,i}\}, x_t, z_t$ ):  
2:      for all cells  $m_i$  do  
3:          if  $m_i$  in perceptual field of  $z_t$  then  
4:               $l_{t,i} = l_{t-1,i} + \text{inverse\_sensor\_model}(m_i, x_t, z_t) - l_0$   
5:          else  
6:               $l_{t,i} = l_{t-1,i}$   
7:          endif  
8:      endfor  
9:      return  $\{l_{t,i}\}$ 
```

Occupancy Grid Mapping

- Log odds representation :
 - 0 ve 1 civarı doluluk olasılığındaki sayısal stabilite

$$\ell_{t,i} = \log \frac{p \left(\mathbf{m}_i \mid z_{1:t}, x_{1:t} \right)}{1 - p \left(\mathbf{m}_i \mid z_{1:t}, x_{1:t} \right)}$$

Occupancy Grid Mapping

- Log odds representation → hücre doluluk olasılığı

$$p \left(\mathbf{m}_i \mid z_{1:t}, x_{1:t} \right) = 1 - \frac{1}{1 + e^{\ell_{t,i}}}$$

Occupancy Grid Mapping

- Log odds representation ilklendirme

$$\ell_0 = \log \frac{p(\mathbf{m}_i = 1)}{p(\mathbf{m}_i = 0)} = \log \frac{p(\mathbf{m}_i = 1)}{1 - p(\mathbf{m}_i = 1)} = 0$$

Occupancy Grid Mapping

- Inverse Sensor Model

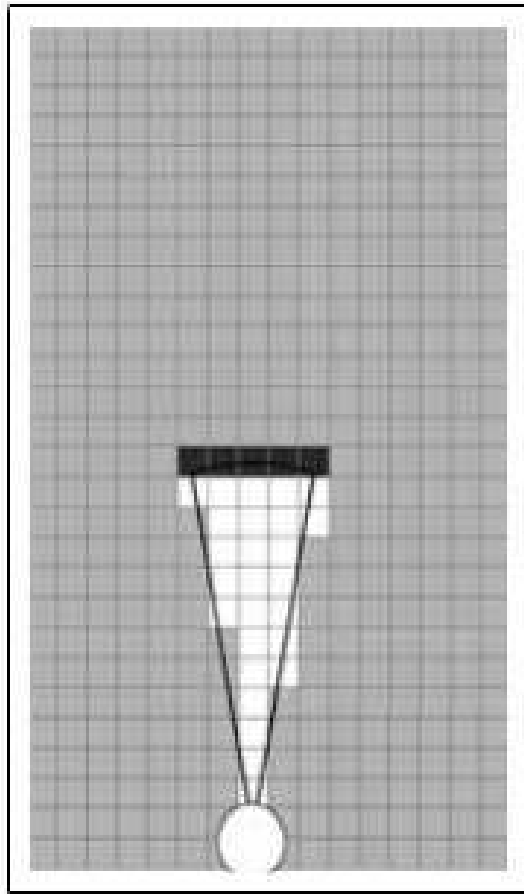
$$\text{inverse sensor model } (\mathbf{m}_i, x_t, z_t) = \log \frac{p \left(\mathbf{m}_i \mid z_t, x_t \right)}{1 - p \left(\mathbf{m}_i \mid z_t, x_t \right)}$$

Occupancy Grid Mapping

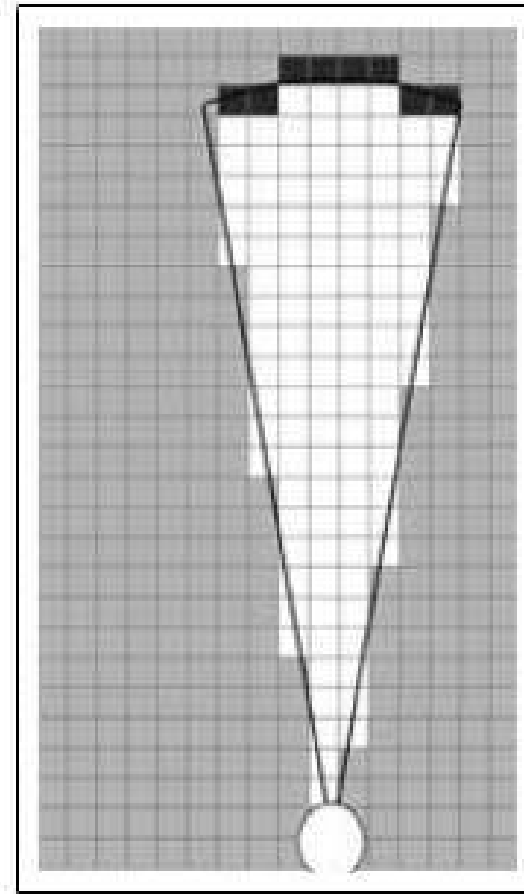
```
1:   Algorithm inverse_range_sensor_model( $\mathbf{m}_i, x_t, z_t$ ):
2:       Let  $x_i, y_i$  be the center-of-mass of  $\mathbf{m}_i$ 
3:        $r = \sqrt{(x_i - x)^2 + (y_i - y)^2}$ 
4:        $\phi = \text{atan2}(y_i - y, x_i - x) - \theta$ 
5:        $k = \text{argmin}_j |\phi - \theta_{j,\text{sens}}|$ 
6:       if  $r > \min(z_{\text{max}}, z_t^k + \alpha/2)$  or  $|\phi - \theta_{k,\text{sens}}| > \beta/2$  then
7:           return  $l_0$ 
8:       if  $z_t^k < z_{\text{max}}$  and  $|r - z_t^k| < \alpha/2$ 
9:           return  $l_{\text{occ}}$ 
10:      if  $r \leq z_t^k$                                       $l_{\text{occ}} > l_0$ 
11:          return  $l_{\text{free}}$                                 $l_{\text{free}} < l_0$ 
12:      endif
```

Occupancy Grid Mapping

(a)



(b)



Occupancy Grid Mapping -Max. A Posteriori

